

Integration — Lesson 19 Test: Initial Values & Net Change

10 questions · show your work in the box · check answers by differentiating and by plugging in the known point · include units in story problems

Name: _____ Date: _____ Score: _____ / 10

1. Given $f'(x) = 2x$ and $f(0) = 5$, find $f(x)$.

1 pt

2. Given $f'(x) = 3x^2 + 2$ and $f(1) = 8$, find $f(x)$.

1 pt

3. A robot rolls at a steady velocity $v(t) = 8$ meters per second and starts at position $s(0) = 2$ meters. Find $s(t)$, then find where the robot is at $t = 3$ seconds.

4. A cart has velocity $v(t) = 2t$ m/s, and at time $t = 2$ its position is $s(2) = 9$ m. Find $s(t)$, then ^{1 pt} find $s(4)$. (Careful: the known point is NOT at $t = 0$.)

5. A tank holds 15 liters at $t = 0$ and fills at a rate of $r(t) = 2 + 4t$ liters per minute. How ^{1 pt} much water is in the tank at $t = 3$ minutes?

6. You start with \$30. You earn money at a rate of $e(t) = 5 + 2t$ dollars per day, where t is in ^{1 pt} days. How much money do you have after 4 days?

7. Suppose $F(2) = 7$ and $\int_2^5 F'(t) dt = 12$. Use the net change theorem to find $F(5)$, and write ^{1 pt} one sentence explaining the idea ("later amount = ...").

8. Velocity $v(t) = 6t$ m/s, starting position $s(0) = 4$ m. Find $s(2)$ TWO ways: (a) build $s(t)$ by finding C , then evaluate; (b) compute $s(0) + \int_0^2 v(t) dt$. Show that both give the same number. 1 pt

9. A tank's amount function comes out to $A(t) = 4t + C$ liters. In plain language, what does the C mean in this story, and what single piece of information would let you find it? 1 pt

10. A plant is 5 cm tall at week $t = 0$ and grows at a rate of $g(t) = 2t + 2$ centimeters per week. (a) Find the height function $h(t)$. (b) In what week does the plant reach exactly 13 cm? 1 pt
